

Advanced Building Permit Management System

1. System Overview

The proposed system is a cloud-based web portal that digitizes the building permit process. Applicants can submit permit applications online, upload documents, track approval progress, and pay permit fees digitally.

The system will support:

- Online application submission
- Document upload and validation
- Permit workflow management
- Online payment via GCash
- Administrative review dashboard
- Status tracking and notifications

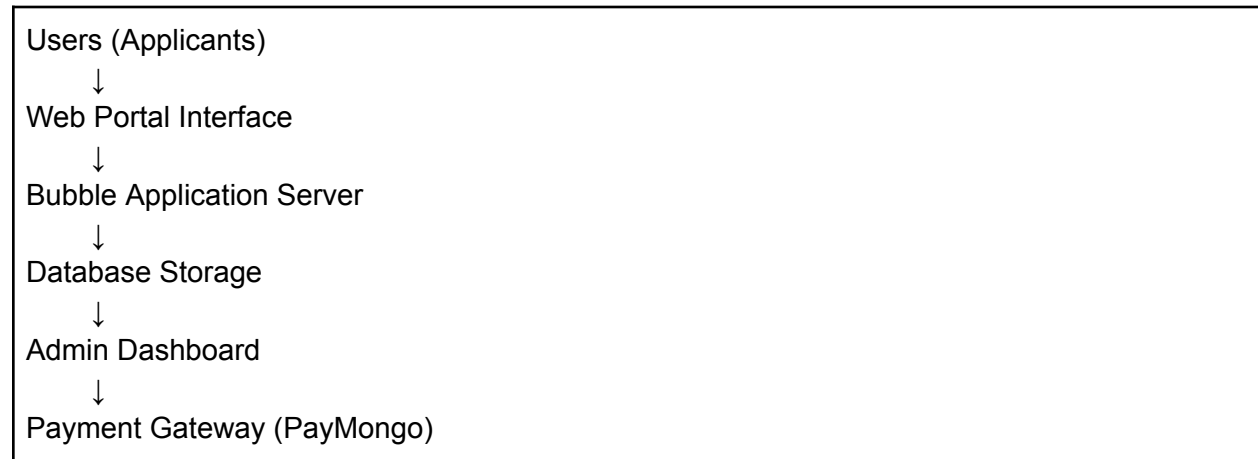
Payments will be integrated using PayMongo, which supports GCash and other digital payments.

2. System Architecture

Technology Stack

Layer	Technology
Frontend	Bubble visual editor
Backend	Bubble database & workflows
Storage	Bubble cloud storage
Payments	PayMongo API
Notifications	Email / SMS API
Hosting	Bubble cloud hosting

System Architecture Diagram



3. User Roles

1. *Public User (Applicant)*

Capabilities:

- Register account
- Submit permit application
- Upload building documents
- Track application status
- Receive notifications
- Pay permit fees

2. *Engineering Office Staff*

Capabilities:

- Review submitted applications
- Verify uploaded documents
- Request additional requirements
- Approve / reject permits
- Generate permit numbers

3. *System Administrator*

Capabilities:

- Manage system settings
- Manage user roles
- Configure permit requirements
- Generate reports

4. System Modules

Module 1: User Registration & Authentication

Features:

- secure login
- account registration
- password encryption
- session management

Registration fields:

- Full Name
- Email Address
- Mobile Number
- Address
- Password
- Confirm Password

Module 2: Permit Application Submission

Applicants submit construction details.

- Application Form Fields
- Application Type
- Project Address
- Barangay
- Lot Area
- Building Floor Area
- Number of Floors
- Construction Type
- Estimated Project Cost
- Architect / Engineer Name
- License Number

Module 3: Document Upload Management

Applicants upload required files.

Required Documents

- Building Plan
- Structural Plan
- Electrical Plan
- Lot Title
- Tax Declaration

- Barangay Clearance
- Fire Safety Clearance
- Contractor License

Accepted file formats: PDF, PNG, JPG

Module 4: Smart File Requirement Checker

Before submission, the system validates:

- ☐ Required documents uploaded
- ☐ Correct file format
- ☐ File size limit

If missing files: Application cannot proceed until complete.

Module 5: Application Workflow Engine

Each permit goes through multiple approval stages.

Workflow Stages

1. Submitted
2. Initial Review
3. Technical Evaluation
4. Compliance Required
5. For Payment
6. Payment Verified
7. Permit Approved
8. Ready for Release

Module 6: Status Tracking System

Applicants can check status through: Application ID

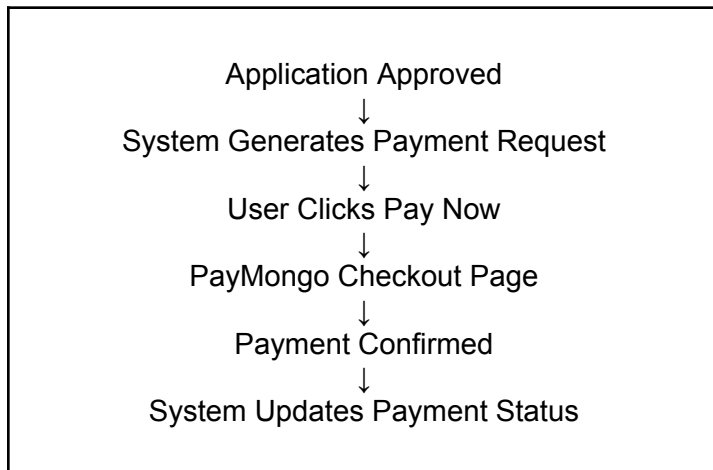
Module 7: Payment System

Payment integration via PayMongo.

Supported payments:

GCash, Maya, Credit cards, Debit cards

Payment Workflow



Payment confirmation is automatically returned to Bubble.

Module 8: Notification System

Notifications are sent when status changes.

Channels:

- Email
- SMS

Module 9: Admin Dashboard

The dashboard provides a centralized control panel for staff.

Dashboard Components

- Application List Table

Application ID	Applicant	Status	Payment	Date
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- Document Viewer
Admins can open uploaded files directly.
- Status Update Controls
Admins can update application status.

Module 10: Analytics & Reporting

Reports available:

- Monthly Applications
- Approved Permits
- Pending Reviews
- Revenue from Permit Fees

Graphs and statistics for management planning may be used.

5. Database Design

Table: Users

- User ID
- Name
- Email
- Mobile
- Role
- Address
- Date Registered

Table: Applications

- Application ID
- User ID
- Project Address
- Permit Type
- Status
- Submission Date
- Payment Status
- Assigned Officer

Table: Payments

- Payment ID
- Application ID
- Amount
- Payment Method
- Transaction ID
- Payment Status
- Payment Date

6. Security Design

Security measures include:

- encrypted login credentials
- secure database storage
- role-based permissions
- protected admin dashboard
- HTTPS encryption

7. User Interface Design

Homepage Layout

Sections:

- City Logo
- Portal Title
- Apply Button
- Status Checker
- Requirements List
- Announcements

Applicant Dashboard

Displays:

- My Applications
- Application Status
- Upload Additional Files
- Payment Link

Admin Dashboard

Displays:

- Application Queue
- Review Documents
- Update Status
- Payment Records
- Analytics Charts

8. Future Enhancements

Possible upgrades include:

- AI document validation
- automatic permit fee calculation
- inspection scheduling module
- digital permit issuance (PDF)
- GIS property mapping

9. Estimated Development Timeline

Phase	Duration
System Design	2 weeks
Development	5 weeks
Testing	2 weeks
Deployment	1-2 weeks

Total Timeline: 12-15 weeks

Final Result

A fully digital building permit management platform that improves:

- processing efficiency
- applicant convenience
- administrative tracking
- revenue transparency